

Incident Management System

MVP — Create → Assign → Investigate → Fix → Verify → Close

Version: 1.0

Scope: MVP

Document purpose: Build-ready product requirements for an internal Incident Management application.

Document decisions incorporated

- MVP fokus pada lifecycle Create → Assign → Investigate → Fix → Verify → Close.
- Pengguna: User/Customer, Help Desk/Support, System Analyst, Developer, QA, DevOps, Manager/Lead.
- Incident dapat berasal dari laporan manual, Help Desk, Email/WhatsApp, maupun sumber otomatis.
- Severity dan Priority dipisahkan: Severity mengukur dampak; Priority mengukur urgensi/kecepatan penanganan.
- Konvensi database: tabel master menggunakan prefix master_; tabel transaksi menggunakan prefix trans_.

1. TL;DR

Incident Management System adalah aplikasi terpusat untuk mencatat, mengelola, memantau, dan menyelesaikan incident yang terjadi pada aplikasi atau layanan perusahaan. MVP memprioritaskan workflow operasional inti dari incident dibuat sampai ditutup.

Core workflow: Create → Assign → Investigate → Fix → Verify → Close.

2. Problem Statement

Incident saat ini dapat dilaporkan melalui berbagai channel seperti Help Desk, email, WhatsApp, atau langsung kepada technical team. Tanpa satu sumber data terpusat, ownership, status, riwayat tindakan, dan hasil penyelesaian sulit ditelusuri.

- Incident berpotensi tidak tercatat secara terstruktur.
- Sulit mengetahui PIC dan status terkini.
- Riwayat investigasi dan fixing tersebar.
- Manager/Lead kesulitan mendapatkan gambaran incident aktif dan workload.
- Data historis sulit digunakan untuk analisis incident berulang.

Problem statement: perusahaan membutuhkan satu sumber informasi terpusat untuk seluruh siklus penanganan incident dari laporan sampai penyelesaian.

3. Goals

Business goals

- Menyediakan centralized incident repository.
- Meningkatkan visibility terhadap incident aktif dan histori.
- Memperjelas ownership dan accountability.
- Mengurangi incident yang terlupakan atau tidak memiliki PIC.
- Menyediakan data untuk monitoring dan reporting.

User goals

- User dapat melaporkan incident dan melihat statusnya.
- Help Desk dapat membuat, melakukan triage, assign, dan memonitor incident.
- System Analyst dapat menganalisis dan mengoordinasikan penyelesaian.
- Developer/DevOps dapat melakukan investigation dan mencatat fixing.
- QA dapat melakukan verification dan mengembalikan incident jika gagal.
- Manager/Lead dapat memonitor incident dan workload.

4. Non-Goals — MVP

- Problem Management penuh.
- Change Management.
- CMDB dan Service Catalog.
- Advanced SLA/on-call scheduling.
- Automatic infrastructure remediation.
- AI Root Cause Analysis dan predictive incident detection.
- Knowledge Base sebagai produk terpisah.

5. Users & Roles

Role	Tanggung jawab utama
User / Customer	Membuat incident, melihat status, memberi informasi tambahan.

Role	Tanggung jawab utama
Help Desk / Support	Menerima laporan, create/triage, assignment, monitoring, komunikasi.
System Analyst	Analisis incident, koordinasi technical team, membantu menentukan resolution.
Developer	Investigasi dan melakukan code/application fixing.
QA	Melakukan verification/testing terhadap hasil fixing.
DevOps	Investigasi infrastructure/deployment dan melakukan remediation.
Manager / Lead	Monitoring incident, workload, dan reporting.

6. Incident Lifecycle

Status MVP:

Status	Definisi	Transisi umum
NEW	Incident baru dibuat.	→ ASSIGNED
ASSIGNED	Team dan PIC sudah ditentukan.	→ INVESTIGATING
INVESTIGATING	PIC melakukan analisis dan mencari penyebab.	→ FIXING
FIXING	Perbaikan sedang dikerjakan/dideploy.	→ VERIFYING
VERIFYING	Fix sedang diuji/diverifikasi.	→ RESOLVED atau FIXING
RESOLVED	Fix sudah berhasil diverifikasi.	→ CLOSED
CLOSED	Incident selesai dan lifecycle ditutup.	Terminal

7. Severity vs Priority

Severity menjawab: “Seberapa besar dampak masalah?” Priority menjawab: “Seberapa cepat masalah harus ditangani?” Keduanya sengaja dipisahkan agar incident berdampak kecil tetapi sangat urgent tetap dapat diberi priority tinggi.

Level	Severity	Priority
1	S1 Critical — dampak sangat besar terhadap layanan/bisnis.	P1 Critical — perlu tindakan segera.
2	S2 High — dampak besar tetapi masih ada sebagian fungsi/workload.	P2 High — perlu ditangani segera setelah P1.
3	S3 Medium — dampak terbatas dan bisnis masih berjalan.	P3 Normal — ditangani sesuai queue/planning.
4	S4 Low — dampak kecil/minor.	P4 Low — dapat dijadwalkan.

Contoh: typo pada halaman login sebelum demo customer dapat memiliki Severity S4 Low tetapi Priority P1/P2 karena urgency bisnisnya tinggi. Sebaliknya, bug besar pada report development dapat memiliki Severity S1/S2 tetapi Priority lebih rendah karena tidak berdampak pada production saat itu.

8. User Stories & Acceptance Criteria

US-01 — Create Incident

Sebagai User/Support, saya ingin membuat incident agar masalah tercatat dan dapat ditindaklanjuti.

- Given user memiliki permission create incident, when field wajib valid dan submit, then sistem membuat incident dengan status NEW.
- Incident mendapatkan nomor unik.
- Created by dan created_at otomatis tersimpan.

US-02 — Assign Incident

- Given incident NEW, when authorized user memilih team dan PIC, then assignment tersimpan dan status menjadi ASSIGNED.
- Assignment change tercatat pada activity history.
- PIC menerima notification.

US-03 — Investigation

- PIC dapat mengubah status menjadi INVESTIGATING.
- Investigation notes dan finding dapat dicatat.
- Perubahan status dan aktivitas tercatat.

US-04 — Fix

- PIC dapat mencatat fix description dan reference deployment/PR bila tersedia.
- Status dapat berpindah ke FIXING lalu VERIFYING.

US-05 — Verification

- QA/authorized verifier memilih PASS atau FAIL.
- PASS → RESOLVED.
- FAIL → FIXING.
- Jika FAIL, reason wajib diisi dan PIC menerima notification.

US-06 — Close

- Authorized user dapat close incident yang sudah RESOLVED.
- closed_at dan closed_by tercatat.
- Incident tidak dapat melanjutkan lifecycle normal setelah CLOSED tanpa reopen action.

9. User Experience / Pages

Page	Fungsi
Login	Authentication.
Dashboard	Summary incident, breakdown status/priority/team/application, my incidents.
Incident List	Search, filter, sort, pagination, quick status/priority visibility.
Create Incident	Form pembuatan incident.
Incident Detail	Detail, assignment, timeline, investigation, fix, verification, comments, attachments.
My Incidents	Incident yang dimiliki/ditugaskan kepada user.
Master Data	Pengelolaan master yang diperlukan dan dibatasi admin/authorized role.

Required states

- Loading: tampilkan loading state/skeleton saat data sedang diambil.
- Empty: jelaskan bahwa belum ada incident dan berikan CTA Create Incident bila user berhak.
- Success: tampilkan confirmation setelah create, assign, update, verification, atau close.
- Error: tampilkan pesan yang dapat ditindaklanjuti dan jangan menghilangkan input yang sudah diisi.
- Recovery: user dapat retry setelah network/server error; status transition tidak boleh terjadi dua kali karena double submit.

10. Functional Requirements

ID	Requirement	Priority
FR-01	Create incident dengan field wajib dan nomor incident unik.	Must
FR-02	Search/filter/sort incident.	Must
FR-03	Assignment team dan PIC.	Must
FR-04	Controlled status transition.	Must
FR-05	Investigation notes.	Must
FR-06	Fix record dan reference.	Must

ID	Requirement	Priority
FR-07	Verification PASS/FAIL.	Must
FR-08	Comment dan activity timeline.	Must
FR-09	Attachment.	Should
FR-10	Dashboard summary.	Should
FR-11	In-app/email notification.	Should
FR-12	Reopen resolved/closed incident dengan permission khusus.	Should

11. Data Model & Naming Convention

Database naming convention: master/reference tables menggunakan prefix master_; transaction/event tables menggunakan prefix trans_.

Master tables

Table	Purpose
master_user	User/account reference.
master_team	Team/department yang menangani incident.
master_application	Daftar aplikasi/service yang dapat dikaitkan dengan incident.
master_environment	Production, staging, development, dan environment lain.
master_incident_source	User, Help Desk, Email, WhatsApp, System/Monitoring, Other.
master_incident_priority	P1-P4.
master_incident_severity	S1-S4.
master_incident_status	NEW, ASSIGNED, INVESTIGATING, FIXING, VERIFYING, RESOLVED, CLOSED.

Transaction tables

Table	Purpose
trans_incident	Core incident record.
trans_incident_assignment	Riwayat team/PIC assignment.
trans_incident_comment	Komentar/komunikasi pada incident.
trans_incident_activity	Audit trail/status transition/activity.
trans_incident_investigation	Investigation notes dan findings.
trans_incident_fix	Fix/action record.
trans_incident_verification	Verification result PASS/FAIL.
trans_incident_attachment	File/attachment terkait incident.

Core trans_incident fields

Field	Description
id	Primary key.
incident_no	Human-readable unique incident number, e.g. INC-2026-000123.
title	Judul singkat incident.
description	Deskripsi masalah.
source_id	FK ke master_incident_source.
severity_id	FK ke master_incident_severity.
priority_id	FK ke master_incident_priority.

Field	Description
status_id	FK ke master_incident_status.
application_id	FK ke master_application.
environment_id	FK ke master_environment.
reporter_id	FK ke master_user.
created_at / updated_at	Audit timestamps.
resolved_at / closed_at	Lifecycle timestamps.

12. Assignment Model

Rekomendasi MVP: satu incident memiliki satu owner/PIC aktif dan satu assigned team, tetapi dapat memiliki beberapa collaborator melalui mekanisme yang dapat ditambahkan kemudian. Ownership tunggal membuat accountability jelas.

13. Notifications

- Incident created.
- Incident assigned/reassigned.
- Status changed.
- New comment.
- Verification failed.
- Incident resolved.
- Incident closed.

MVP channel: in-app notification + email. WhatsApp dapat tetap menjadi source laporan, tetapi integrasi WhatsApp sebagai notification engine diposisikan sebagai pengembangan berikutnya.

14. Dashboard & Reporting

- Total open incident.
- Incident by status.
- Incident by severity.
- Incident by priority.
- Incident by application.
- Incident by team/PIC.
- My open incidents.
- Average time Create → Assign.
- Average time Create → Resolved.
- Average time Create → Closed.
- Reopen rate.
- Verification failure rate.

Jangan menggunakan jumlah closed incident sebagai satu-satunya indikator keberhasilan. Reopen rate dan verification failure rate perlu dipantau untuk menghindari optimasi semu.

15. Technical Considerations

Authentication & authorization

- Authentication mengikuti mekanisme identity yang digunakan organisasi.
- Authorization berbasis role + ownership/team + permission.
- Setiap perubahan lifecycle harus divalidasi server-side.

Audit & security

- Semua status transition dan assignment change harus memiliki actor dan timestamp.
- Attachment perlu validasi tipe/ukuran file dan access control.
- User hanya boleh melihat incident sesuai scope permission.
- Data sensitif pada description/comment perlu mengikuti kebijakan perusahaan.

Performance

- List menggunakan pagination.
- Search/filter menggunakan indexed fields yang relevan.
- Activity timeline tidak boleh menyebabkan query N+1.
- Notification dikirim secara asynchronous bila volume meningkat.

16. API Contract — Initial

Method	Endpoint	Purpose
POST	/api/incidents	Create incident.
GET	/api/incidents	List/search/filter incidents.
GET	/api/incidents/{id}	Incident detail.
PATCH	/api/incidents/{id}/assignment	Assign/reassign.
PATCH	/api/incidents/{id}/status	Controlled status transition.
POST	/api/incidents/{id}/comments	Add comment.
POST	/api/incidents/{id}/investigations	Add investigation record.
POST	/api/incidents/{id}/fixes	Add fix record.
POST	/api/incidents/{id}/verification	Submit PASS/FAIL.
POST	/api/incidents/{id}/close	Close incident.

API error shape should be standardized, for example: code, message, details, request_id. Status transition endpoint must reject invalid transitions server-side.

17. Success Metrics

Category	Metric
Leading	% incident memiliki PIC; % incident memiliki investigation; % incident memiliki verification result.
Outcome	Average Create→Assign; Create→Resolved; Create→Closed; reopen rate; verification failure rate.
Guardrail	Duplicate incident rate; notification failure rate; unauthorized access attempts; stale open incidents.

18. Milestones & Sequencing

Phase	Relative duration	Scope
Phase 1	± 2-3 weeks	Authentication, role/permission, create, list, detail, assignment, status, comment, activity.
Phase 2	± 1-2 weeks	Investigation, fix, verification, resolution, close, reopen.
Phase 3	± 1-2 weeks	Dashboard, reporting, notification, advanced search/filter.

Durasi adalah estimasi sequencing untuk membantu planning, bukan commitment delivery.

19. Risks, Open Questions & Decisions

Topic	Decision / Open question
Application master	Incident MVP sebaiknya dikaitkan dengan master_application agar reporting dapat dikelompokkan per aplikasi/
Severity/Priority	Dipisahkan. Final mapping matrix perlu disepakati oleh business/IT operations.

Topic	Decision / Open question
PIC model	Satu owner/PIC aktif per incident; collaborator dapat dikembangkan kemudian.
SLA	Belum menjadi core MVP; struktur data sebaiknya tidak menghambat penambahan SLA di fase berikutnya.
External sources	Email/WhatsApp/system automation dapat dibuat sebagai integration layer setelah core workflow stabil.

20. Coding-Agent Handoff

Scope

- Implement incident lifecycle dan controlled state transition.
- Implement role/permission sesuai matrix.
- Implement database master_/trans_ sesuai data model.
- Implement incident list/detail/create/assignment/investigation/fix/verification/close.
- Implement audit activity dan basic notification.

Modules/files to inspect

- Authentication/authorization module.
- Existing user/team/application master modules.
- Database migration/schema conventions.
- API routing/controller/service/repository conventions.
- Notification service jika sudah tersedia.

Acceptance test matrix

- Create valid incident → NEW.
- Assign team/PIC → ASSIGNED.
- Start investigation → INVESTIGATING.
- Record fix → FIXING/VERIFYING.
- Verification PASS → RESOLVED.
- Verification FAIL → FIXING dan failure reason wajib.
- Close RESOLVED → CLOSED.
- Unauthorized transition → 403/appropriate authorization error.
- Invalid transition → validation error dan status tidak berubah.
- Double submit → tidak membuat duplicate lifecycle action.

Definition of Done

- All Must requirements implemented.
- Database migrations and indexes reviewed.
- API validation and authorization implemented server-side.
- Unit/integration tests cover lifecycle transitions.
- Critical error and recovery states handled.
- Audit trail available for status and assignment changes.
- QA acceptance scenarios pass.
- Basic deployment/configuration documentation available.

21. Recommended Future Enhancements

- SLA and escalation.
- Problem Management.
- Change Management.
- Knowledge Base.

- Monitoring/observability integration.
- Slack/Teams/WhatsApp integrations.
- AI-assisted categorization, similar-incident search, troubleshooting suggestions, and incident summarization.